

FITNESS ACTIVITY PREDICTION

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PROGRAM ABSTRACT:

- This project delves into the classification of human physical activities using sensor data, aiming to enhance the understanding and monitoring of daily fitness routines. Leveraging a comprehensive dataset comprising accelerometer, gyroscope, magnetometer, and heart rate readings, collected across various activities including Running, Squats, Walking, Cycling, and Standing, this study employs a robust machine learning pipeline. The methodology involves thorough Exploratory Data Analysis (EDA) to discern data distributions, relationships, and potential patterns, followed by essential data preprocessing steps such as feature scaling and encoding of target variables. Multiple classification algorithms, including Logistic Regression, Decision Trees, Random Forests, K-Nearest Neighbors, and Support Vector Machines, were trained and rigorously evaluated. Hyperparameter tuning was applied to optimize the performance of the most promising models. The findings demonstrate the efficacy of machine learning in accurately identifying distinct fitness activities from raw sensor data, paving the way for advanced applications in personalized fitness tracking and health management. This research was conducted by Abhimanue R Pillai